

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

Ever wondered how spiders weave their webs? Machine learning has the answer in Ignorance. In Health, learn about the many uses of MRNA, in addition to just vaccinations for COVID-19



Earth's inner secret revealed.

Earth's inner core which includes iron and other lighter elements may be in a supersonic state. This means that the inner core consists of a liquid outer core which covers the solid inner core forming an amalgam of liquid and solid state. The elements forming these cores are still unclear which leaves countless questions about the core secrets of earth.

WHAT'S NEW

Tracking Plastic Pollution in the Ocean from Space

An average of 10 million plastic waste enters the ocean every year. Satellite tracking and mapping of these plastic wastes can prove useful to clean the ocean. The European Space Agency (ESA) and the Dutch institute, Deltares, are currently testing satellite technology that could help track plastic pollution in the ocean. Over 30 promising ideas from 50 ideas were chosen for demonstration purposes and were invited to the facility at Deltares research institute in Delft in the Netherlands. The test campaign was conducted in the Atlantic Basin Facility which is 75 meters long and 9 meters wide

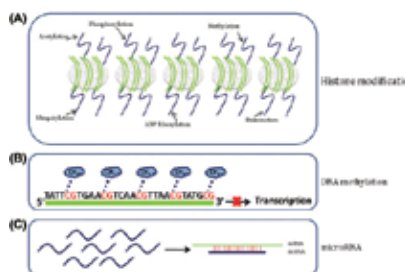


resembling half the size of an Olympic swimming pool. The waves were generated to recreate realistic deep ocean waves. It was conducted

successfully as the instruments were able to detect plastic and the next step would be to take the research outdoors and try it out in a real reservoir. If successfully, the plastic waste hidden deep in the ocean bodies can be tracked and the environment can be preserved.

Cervical cells can be used to identify Ovarian Cancer

Epithelial ovarian cancer, a prevalent type of ovarian cancer, is found in nine out of ten ovarian cancer cases. Most cases go undetected until their final stage. Early detection can save a life. A recent study has shown that cells acquired while testing for cervical cancer could be used to detect Ovarian cancers. Researchers studied 242 women diagnosed with ovarian cancer and 869



without the cancer cells. They studied the epigenetic modifications, molecular variations that changed their gene pattern without destructing the

DNA. The study concluded that the utilisation of epigenetic signatures of DNA methylation will help in the early detection and prevention of ovarian and breast cancers. Additional

research needs to be done to prove the accuracy of these tests but this groundbreaking discovery could potentially save countless lives.



Sunlight plays a role in cleaning up oil spills

Rays of sunlight when shining on the spilt oil can trigger chemical reactions which transform the oil into a new compound and help oil dissolve in the water. The results are still unclear and the impact of these compounds on marine life is not known.



Pluto's atmospheric pressure

A team of astronomers were successful in deriving the value of Pluto's atmospheric pressure at its surface. It is 80,000 times less than the atmospheric pressure at mean sea level on Earth. Scientists from the Aryabhata Research Institute used India's largest Optical telescope for this.



Quantum Boomerang effect

A study confirmed that after giving a nudge, a particle returns to its starting point on average. The catch is that particles can boomerang if they are in a material that has atoms missing or scattered. The boomerang runs on time-reversal symmetry.